

Improved Scalable Analytical Model for Predicting SPM Motor Performance Considering Magnet Shape and Magnetization Pattern

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Abstract—This paper develops the improved scalable analytical model (iSAM) for predicting the performance of the surface-mounted permanent-magnet (SPM) motor with the arbitrary magnet shape and magnetization direction. It extends the ability of the conventional analytical models. The database of the analytical model can store the electromagnetic information for the SPM motors, which enriches the scaling laws for electrical machines. As for the magnets with arbitrary shapes, they are divided into finite rectangles or fan-shaped segments based on their magnetization directions, which are then replaced by equivalent magnet currents along the magnet surfaces. Then, based on the iSAM, the air-gap field of the SPM motor can be obtained considering the iron nonlinearity. This method is beneficial for accelerating the optimization process aimed at reducing torque ripple in servo motor applications. The finite-element method (FEM) for eccentric magnetic pole servo motors is executed to verify the effectiveness of the proposed method. Additionally, a prototype with parallel magnetized PM is tested to demonstrate the accuracy of the iSAM. The proposed method is particularly useful for the modeling techniques and fault diagnosis of electrical machines.

Index Terms—Scalable analytical model, SPM motor, magnet shape, conformal mapping.

I. INTRODUCTION

THE permanent magnet (PM) motor has been widely used in industrial robots due to its high torque density and high efficiency [1]–[4]. The shape and magnetization pattern of PMs show significant influence on motor performance, which is one of the keys to motor optimization [5]. Even though the finite-element method (FEM) shows excellent accuracy in predicting the performance of PM motors with arbitrary magnet shapes [6], [7], it is time-consuming compared with the analytical model.

The magnetic circuit model is challenging in dealing with the irregular magnet shape. Therefore, there are few papers focusing on this topic. In contrast, most papers have only investigated interior PM motors with rectangular magnets [8] or SPM motors with fan-shaped magnets [9] using the magnetic circuit model. The analytical model that directly solves the air-gap field based on Poisson's equation has been widely investigated considering the magnet shape. It has excellent computational

efficiency, but it is only suitable for predicting the magnetic field in the regular domain. The magnetic field of the SPM motors with fan-shaped magnets was predicted using either conformal mapping [10] or subdomain technique [3]. In [11], the equivalent magnetization current theory was presented to calculate the flux-reversal permanent-magnet motors with fan-shaped magnets. The conformal mapping is more capable of dealing with irregular shapes while requiring numerical calculation of some parameters compared with the subdomain technique.

The conventional analytical models combined with the segmentation technique were employed to consider the PMs with arbitrary shape [12]–[15]. In [12]–[14], the subdomain technique combined with either surface current density [12] or segmented magnet pieces [13], [14] was proposed to account for the irregular magnet shape. However, the segmentation technique requires computation of large matrix equations in the subdomain models at each step for all the segmented magnet pieces or surface current density, leading to increased computational burden. In [15], the slotting effect is represented by the surface currents on the stator bore and meanwhile the magnets are segmented into small pieces with rectangular shapes to analyze the slotted SPM motor with different rotor frames and magnet shapes. It requires iterative steps to compute equivalent surface currents for slotting effect and increases computation burden. Furthermore, magnet defect modeling provides an alternative approach to investigate arbitrary magnet geometries [16], [17]. However, the analyzed magnets remain geometrically regular, sharing identical inner and outer diameters. In [18], the expressions of the equivalent PM current were manipulated to facilitate the implementation of the subdomain technique. However, the iron nonlinearity is neglected in [12]–[16], [18], which is the major drawback of these linear analytical models. Table I lists the advantages of the proposed iSAM over those typical analytical models.

The magnetic equivalent circuit is generally unsuitable for SPM motors, due to the inappropriate assumption of lumped magnetic reluctance and localized flux paths in the magnetic equivalent circuit for the motors with large effective air-gap lengths. Very few works have successfully applied magnetic equivalent circuit to SPM motors with great accuracy [19], [20]. In [19], [20], the conventional SPM motors were “meshed” using enormous number of magnetic reluctances, making it similar to FEM, but with a large computational burden. Hence, the magnetic circuit model for SPM motors is not included in this paper.

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The magnetic circuit-analytical hybrid models are widely investigated to consider iron nonlinearity recently. Through subdomain solutions formulated within the iron region, it can analyze spoke-type IPM machines [21]–[23]. Meanwhile, several hybrid approaches combining magnetic-circuit models with subdomain or conformal-mapping techniques have been proposed to evaluate the performance of saturated SPM motors [24], [25]. In [24], the subdomain technique combined with magnetic circuit is employed to analyze the SPM motor with iron nonlinearity, but the magnet shape is limited to rectangular shape with parallel magnetization. In [25], the analytical model combined with the simplified BH iron curve and the magnetic circuit was proposed to predict the SPM motor performance with great accuracy and fast calculation speed. However, these models fail to account for the influence of the magnet shape and magnetization pattern on the motor performance, which is one of the significant contributions in this paper. Additionally, this paper is an extension of the conference paper [26].

TABLE I: Comparison of iSAM and Typical Analytical Models

Feature	[10]	[12], [18]	[24]	iSAM
Accuracy	Low	Moderate	Moderate	High
Iron Nonlinearity	No	No	Yes	Yes
Computation	Light	Moderate	Moderate	Moderate
Magnet Shape	Regular	Arbitrary	Regular	Arbitrary
Magnetization Pattern*	Radial	Both	Parallel	Both

* Magnetization pattern includes radial and parallel magnetization.

From an academic perspective, the motivation behind this work is to overcome the limitations of existing motor modeling techniques, as shown in Fig. 1. Finite element method (FEM) offers high accuracy but is computationally intensive and impractical for iterative design or real-time control. Linear analytical model (LAM) is efficient but typically lacks accuracy due to oversimplifications and failure to capture nonlinear effects. Magnetic circuit model (MCM) struggles with SPM motor geometries and becomes inefficient when a large magnetic reluctance matrix is present.

To bridge the gap between computational efficiency and accuracy, this paper proposes the improved scalable analytical model to predict the performance of the surface-mounted permanent-magnet (SPM) motor accounting for the magnet shape and magnetization direction. The superposition principle is used to show the relationship between the total air-gap field and the equivalent current, which includes the winding current, equivalent magnet current, and iron saturation current. Such relationship remains linear, as long as the motor geometry and magnet property are specified. The magnets are finitely divided into rectangles or fan-shaped pieces and then replaced by the equivalent magnet currents along the surface of the magnets based on the superposition principle. The winding current will also be obtained from the winding layout and input current source. However, it is noted that the iron saturation current varies nonlinearly with different input currents due to the nonlinear iron property, leading to the PM motor's nonlinear behavior. Hence, the iSAM can predict the air-gap field of the SPM motor considering slot effect and iron saturation with excellent computational efficiency. The finite-element method (FEM) for eccentric magnetic pole servo motors is executed to

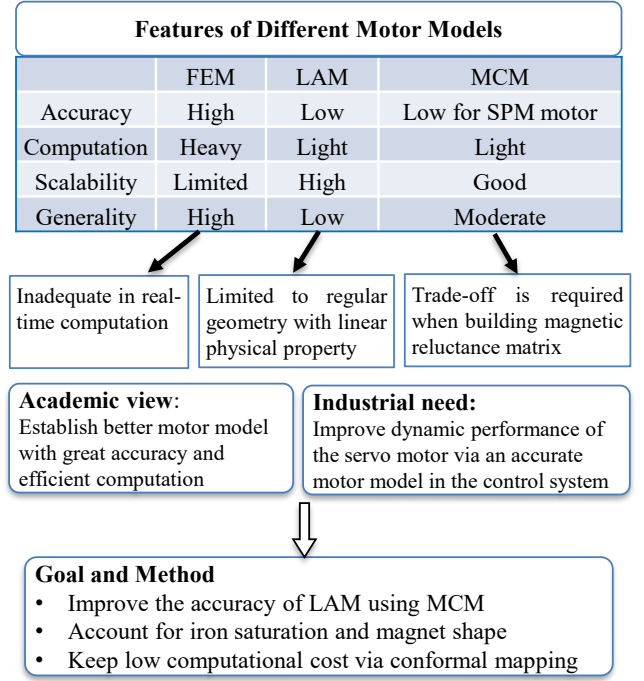


Fig. 1: Motivation of the proposed iSAM

verify the effectiveness of the proposed method. The prototype with parallelly magnetized PM is tested to show the accuracy of iSAM.

The main contribution of the paper is the development of iSAM for predicting the performance of SPM motors:

- 1) Handles arbitrary magnet shapes and magnetization patterns, including eccentric pole-arc shapes and both radial and parallel magnetizations.
- 2) Incorporates iron nonlinearity using equivalent saturation currents with the aid of magnetic circuit, which significantly improves computational accuracy compared with linear analytical models.
- 3) Build the database of the general motors based on conformal mapping technique, which reduces the computational burden compared with FEM and ensures scalability for different SPM motors without numerical calculation.
- 4) Demonstrate high accuracy (average torque errors < 4.5%) and largely reduced computation (time reduction > 90%), which is validated by FEM and experimental results.

The structure of the paper is given as: Section II gives the transformation between the equivalent magnet current and general magnet shape considering magnetization pattern. Then, Section III(A) describes the relationship between the air-gap field distribution and the equivalent current using the iSAM. Based on this theory, the iron saturation current is introduced to account for the motor nonlinearity in Section III(B). The solving process of the proposed method is accordingly presented in Section III(C). Finally, the case study is performed to validate the iSAM in Section IV.

II. GENERAL MAGNET SHAPE

Fig. 2 illustrates the equivalent transformation process from the magnet segment to its corresponding equivalent magnet

current representation based on Ampere's law. The eccentric magnetic pole is circumferentially segmented into small elements, each approximated as a rectangular segment (Fig. 2(a1)-(a2)) for parallel magnetization or a fan-shaped segment (Fig. 2(b1)-(b2)) for radial magnetization.

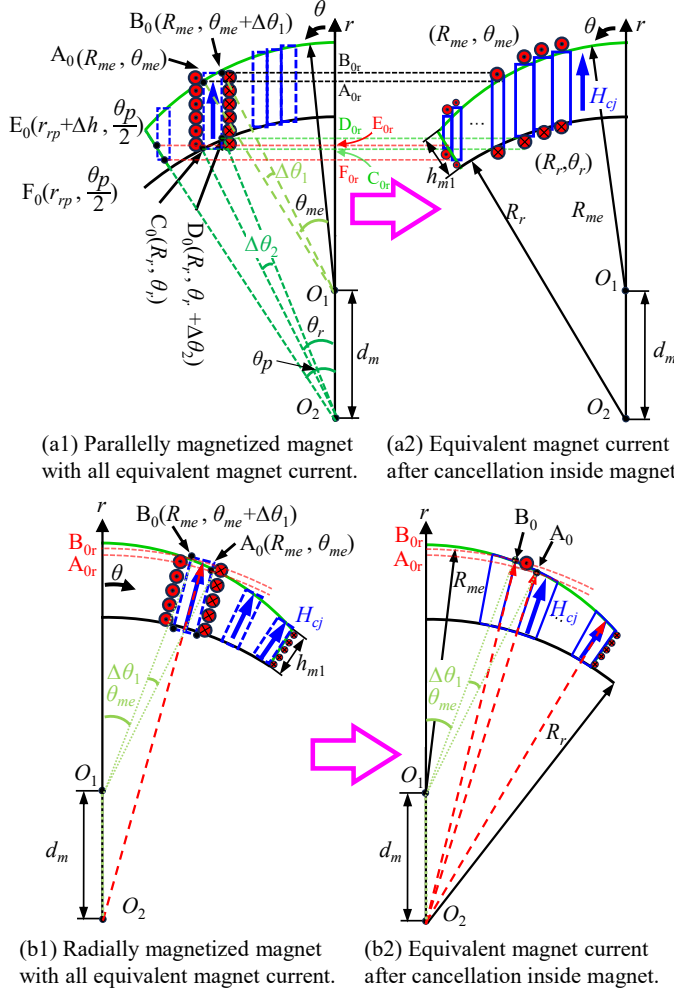


Fig. 2: The transformation from the magnet to the equivalent magnet current.

For parallel magnetization, the permanent magnets are divided into small rectangular elements, as illustrated in Fig. 2(a1)-(a2). The two sides of each rectangular element are aligned with the magnetization direction, and the equivalent surface magnet currents appear at these two sides. Since the complete PM pole may consist of multiple magnet segments, the equivalent currents within adjacent segments cancel each other based on Ampere's law and the principle of magnetization vector superposition, leaving only the currents distributed along the outer and inner magnet surfaces, as shown in Fig. 2(a2). Hence, when discretizing the eccentric magnetic pole into small rectangular elements, the extra PM length is observed, i.e., $|A_{0r}B_{0r}|$ and $|C_{0r}D_{0r}|$ in Fig. 2(a1), to represent the physical arc length corresponding to each equivalent current element. Since $|A_{0r}B_{0r}| = R_{me}[\cos(\theta_{me} + \Delta\theta_1) - \cos(\theta_{me})]$ and $|C_{0r}D_{0r}| = R_r[\cos(\theta_r + \Delta\theta_2) - \cos(\theta_r)]$ according to the geometrical relationship, the equivalent magnet current along the outer and inner surfaces can be expressed

as

$$i_{PMo} = \frac{B_r}{\mu_0} |A_{0r}B_{0r}| \quad (1)$$

$$= \frac{B_r}{\mu_0} R_{me} [\cos(\theta_{me} + \Delta\theta_1) - \cos(\theta_{me})]$$

$$i_{PMi} = -\frac{B_r}{\mu_0} |C_{0r}D_{0r}| \quad (2)$$

$$= -\frac{B_r}{\mu_0} R_r [\cos(\theta_r + \Delta\theta_2) - \cos(\theta_r)]$$

where $|A_{0r}B_{0r}|$ and $|C_{0r}D_{0r}|$ are the projections of arcs A_0B_0 and C_0D_0 along the magnetization direction, respectively. R_{me} and R_r are the outer and inner radii of the magnet. B_r is the residual flux density, μ_0 is the vacuum permeability. $\Delta\theta_1$ and $\Delta\theta_2$ denote the angular widths of the magnet segments measured from the outer and inner centers O_1 and O_2 , respectively.

The PM length per current l_{ms} is defined as the equivalent length of the magnet segment, which will be replaced by equivalent current dot for the analytical model. According to (1)-(2) and Fig. 2(a1), the PM lengths per current for the magnet outer and inner surfaces with parallel magnetization are defined as $l_{ms} = |A_{0r}B_{0r}| = R_{me}[\cos(\theta_{me} + \Delta\theta_1) - \cos(\theta_{me})]$ and $l_{ms} = |C_{0r}D_{0r}| = R_r[\cos(\theta_r + \Delta\theta_2) - \cos(\theta_r)]$, respectively.

At the two lateral sides of the eccentric magnetic pole, $|E_{0r}F_{0r}|$ represents the extra PM height of each rectangular element after cancellation of the equivalent PM currents inside the magnet, as shown in Fig. 2(a1). Since $|E_{0r}F_{0r}| = \Delta h \cos(\theta_p/2)$ from Fig. 2(a1), the corresponding equivalent magnet current at two sides of the eccentric magnetic pole is expressed as

$$i_{PM} = \frac{B_r}{\mu_0} |E_{0r}F_{0r}| \quad (3)$$

$$= \frac{B_r \Delta h}{\mu_0} \cos\left(\frac{\theta_p}{2}\right) = \frac{B_r h_{m1}}{\mu_0 n_0} \cos\left(\frac{\theta_p}{2}\right)$$

where θ_p is the pole-arc angle. $\Delta h = |E_0F_0|$. h_{m1} is the magnet height. n_0 is the number of the uniformly segmented magnets at each side of the eccentric magnetic pole. Accordingly, from (3) and Fig. 2(a1), the PM length per current for the two sides of the eccentric magnetic pole with parallel magnetization is expressed as $l_{ms} = |E_0F_0| = h_{m1}/n_0 \cos(\theta_p/2)$.

For radial magnetization, the magnets are divided into small fan-shaped elements in the tangential direction, as shown in Fig. 2(b1)-(b2). According to Ampere's law and superposition principle of magnetization vectors, no equivalent surface current exists along the inner surface, and the equivalent magnet current along the outer surfaces is calculated using the extra height of the small fan-shaped elements, i.e., $|A_0O_2| - |B_0O_2|$ in Fig. 2(b1). Since $|A_0O_2| = |A_{0r}O_2|$ and $|B_0O_2| = |B_{0r}O_2|$, the equivalent magnet current along the outer surface with radial magnetization is derived as:

$$\begin{aligned}
 i_{PMo} &= -\frac{B_r}{\mu_0}(|A_{0r}B_{0r}|) = -\frac{B_r}{\mu_0}(|A_{0r}O_2| - |B_{0r}O_2|) \\
 &= -\frac{B_r}{\mu_0}(|A_0O_2| - |B_0O_2|) \\
 &= -\frac{B_r}{\mu_0}(|\overrightarrow{O_2O_1} + \overrightarrow{O_1A_0}| - |\overrightarrow{O_2O_1} + \overrightarrow{O_1B_0}|) \quad (4) \\
 &= -\frac{B_r}{\mu_0}(|d_m + R_{me}e^{j\theta_{me}}| - \\
 &\quad |d_m + R_{me}e^{j(\theta_{me}+\Delta\theta_1)}|)
 \end{aligned}$$

where $A_0(R_{me}, \theta_{me})$ and $B_0(R_{me}, \theta_{me} + \Delta\theta_1)$ are points defined in the global polar coordinate system. $d_m = |\overrightarrow{O_2O_1}|$ denotes the eccentricity between the global origin O_2 and local origin O_1 . Similarly, from (4) and Fig. 2(b1), the PM length per current for the magnet outer surfaces with radial magnetization is expressed as $l_{ms} = |A_{0r}B_{0r}| = (|d_m + R_{me}e^{j\theta_{me}}| - |d_m + R_{me}e^{j(\theta_{me}+\Delta\theta_1)}|)$.

Similarly, for the two lateral sides of the eccentric magnetic pole with radial magnetization, the equivalent magnet current is obtained from the PM height of the fan-shaped elements at both side.

$$i_{PM} = \frac{B_r h_{m1}}{\mu_0 n_1} \quad (5)$$

where n_1 represents the number of the equivalent current at each side of the eccentric magnetic pole. Hence, according to (5) and Fig. 2(b1), the PM length per current for the two sides of the eccentric magnetic pole with radial magnetization is expressed as $l_{ms} = h_{m1}/n_1$.

III. SCALABLE ANALYTICAL MODEL

The following assumptions are made in this paper:

- 1) The BH curve of the electrical steel is simplified as the combination of two lines to account for iron nonlinearity.
- 2) The eddy current reaction and end effect are neglected.
- 3) The relative permeability of the magnet is 1 and the property of the magnet is linear.
- 4) The interface between the rotor and magnet should be a circle or arc for employing the same database, while other magnet surfaces can be arbitrary.

A. General analytical model

The analytical model in [25] can only predict the SPM motor performance with fan-shaped radially magnetized magnets. This paper extends the ability of the iSAM to consider the magnet shape (eccentric magnet pole in this paper), magnetization direction, and iron nonlinearity, while keeping high computational efficiency. In iSAM, only the general motor with variable slot-opening and air-gap length is required to predict the new motor's performance. The conformal mapping is used to build the relationship between the slotted air-gap field and the equivalent current, including the winding current, equivalent magnet current, and the saturation current of iron, which is extracted from the database of the general motor. In Fig. 3, only transformation is used to get the same shape between the general motor and new motor. Then, the

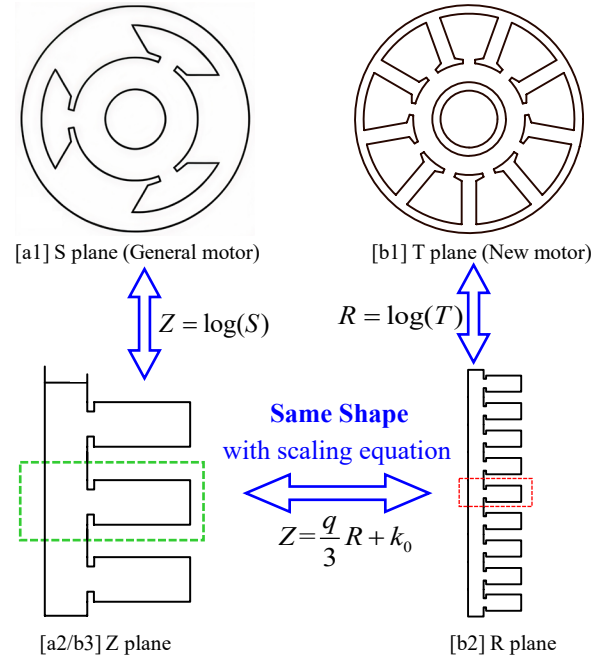


Fig. 3: The conformal mapping relationship between the new and general motor.

linear scaling function is proposed to build the mathematical connection between [a2/b3] and [b2].

Since the relationship between the new motor and the general motor is revealed using Fig. 3, it is feasible to consider obtaining the magnetic potential in the general motor first and conformally transform to the new motor with reduced computation. Therefore, the database of the general motor is required based on the conformal mapping method, as shown in Fig. 4. To consider the slotting effect, the logarithmic transformation is firstly used to obtain the polygonal shape of the air-gap region from the slotted air-gap according to Fig. 4(i)-(ii). Then the Schwarz-Christoffel transformation is applied to transform the polygonal shape into the rectangle [27]–[29], as shown in Fig. 4(ii)-(iii). Finally, the exponential function is used to transform the rectangular air-gap into the slotless air-gap in Fig. 4(iii)-(iv). The parameters in these functions can be numerically obtained based on the geometry of the general motor and the positions of the current and magnetic potential. For the database of the general motor, only the slot opening length and the total equivalent air-gap length (including the air-gap length and the PM height) are required to select the corresponding magnetic potential for the new motor. Therefore, it is not difficult to build this database even using ordinary computational resources. Besides, to reduce the complexity of the conformal mapping process (logarithm function) in Fig. 4(i)-(ii) while keeping the generality of the proposed method, it is feasible to refine the geometry of the rotor-magnet interface to be circular, which means that only two points are required to define the rotor-magnet interface after conformal mapping. This restriction is reasonable since most SPM motors have circular rotor-magnet interfaces. Nevertheless, it is possible to extend the method to represent the rotor-magnet interfaces with arbitrary shapes by using more complex points along the rotor-magnet interface

during the conformal mapping process.

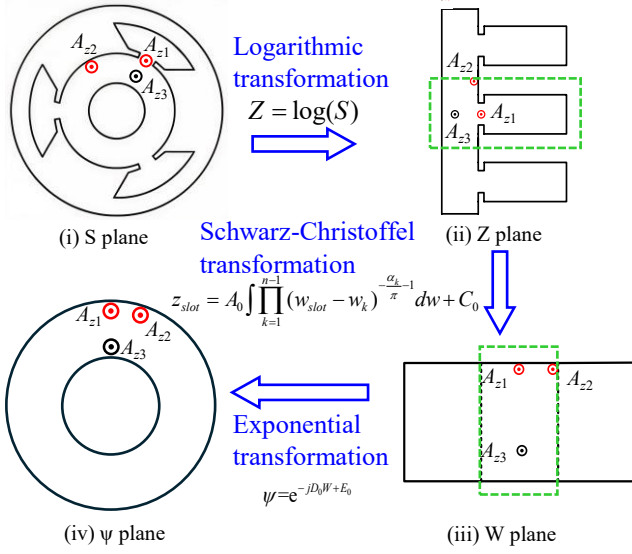


Fig. 4: The conformal mapping process to extract the magnetic potential in the general motor.

The total air-gap field is produced by the PMs, the winding current, and the equivalent iron saturation current.

$$\begin{aligned} B_r &= \sum_{k=1}^{N_k} B_{ri}(i_{PMk}) + \sum_{m=1}^{N_m} B_{ri}(i_w) + \sum_{n=1}^{N_n} B_{ri}(i_{sat}) \\ B_\theta &= \sum_{k=1}^{N_k} B_{\theta i}(i_{PMk}) + \sum_{m=1}^{N_m} B_{\theta i}(i_w) + \sum_{n=1}^{N_n} B_{\theta i}(i_{sat}) \end{aligned} \quad (6)$$

where i_{PMk} , i_w , and i_{sat} are the equivalent magnet current, winding current and iron saturation current while N_k , N_m , and N_n are the total number of these equivalent currents, respectively. The radial and tangential air-gap flux density $B_{ri}(i_k)$ and $B_{\theta i}(i_k)$ produced by the dot current i_k in the slotted air-gap is derived from the analytical solution of vector potential $A_k(i_k)$ in the slotless air-gap combined with the conformal mapping method [25]

$$\begin{aligned} B_{ri}(i_k) + jB_{\theta i}(i_k) &= \lambda \left(\frac{1}{r} \frac{\partial A_k(i_k)}{\partial \theta_\varphi} - j \frac{\partial A_k(i_k)}{\partial r_\varphi} \right) \\ A_k(i_k) &= \frac{\mu_0 i_k}{2\pi} \left\{ \ln \sqrt{r_\varphi^2 + r_{k\varphi}^2 - 2r_\varphi r_{k\varphi} \cos(\theta_\varphi - \theta_{k\varphi})} - \sum_{i=1}^{+\infty} \frac{r_\varphi^i r_{s\varphi}^{-i} (r_{k\varphi}^{2i} + r_{r\varphi}^{2i}) + r_\varphi^{-i} r_{r\varphi}^{2i} (r_{s\varphi}^{-i} r_{k\varphi}^{2i} + r_{s\varphi}^i)}{i (r_{s\varphi}^{2i} - r_{r\varphi}^{2i})} \right. \\ &\quad \left. \cdot \cos(i(\theta_\varphi - \theta_{k\varphi})) \right\} \end{aligned} \quad (7)$$

where the current position $(r_{k\varphi}, \theta_{k\varphi})$, magnetic field position $(r_\varphi, \theta_\varphi)$, and the complex permeance function λ from the conformal mapping in Fig. 4 are all stored in the database of the general motor. i_k is the equivalent current, including the winding current, equivalent magnet current, and the saturation current of iron. $r_{s\varphi}$ and $r_{r\varphi}$ are the outer and inner radii of the annulus in Ψ plane of Fig. 4. As $B_{ri}(i_k)$ and i_k are linearly related according to (7)-(8), it is feasible to predict the

total air-gap field using superposition principle. Meanwhile, the nonlinear values of i_{sat} account for the saturation level in the PM motor and the calculation process will be given in the next subsection.

The generation time of the iSAM database was approximately 3 days on a standard engineering workstation (Intel Core i7, 64GB RAM), and it is only required once for a set of general motor configurations. Since the database is reusable and parameterized by only slot opening and equivalent air-gap length, it is well-suited for practical engineering workflows where multiple design variations are explored. To be more specific, the extraction of the parameters for the conformal mapping [28]-[29] and the positions between S plane and Ψ plane in Fig. 4 are all replaced by the database of the general motor to accelerate the computational efficiency.

B. Equivalent iron saturation current

The iron saturation is also represented by the equivalent current along the surfaces of the stator yoke, tooth body, and rotor back iron, with the help of the magnetic circuit [24], [26], [29]. To simplify the iSAM, the iron saturation current representing the stator yoke and tooth body will be placed at the center of the slot. Such simplification only introduces minor errors as the differences of the magnetic potential in the whole slot region are usually small for PM motors [30]-[32]. Fig. 5 illustrates the justification of replacing the iron nonlinearity with the equivalent saturation current i_{sat} based on Ampere's law. Along the dashed loop with finite iron permeability in Fig. 5(a), the following equation can be obtained

$$F_{tooth1} + F_{yoke} + F_{tooth2} + \int_{L_{airgap}} H_{airgap} dl = i_w \quad (9)$$

where F_{tooth1} , F_{tooth2} , and F_{yoke} are the magnetic potential drops in tooth body 1, tooth body 2, and stator yoke, respectively.

Then, for the equivalent model of Fig. 5(b), Ampere's law can give:

$$0 + \int_{L_{airgap}} H_{airgap} dl = i_w - i_{sat} \quad (10)$$

To keep the same air-gap field in Fig. 5, the value of the equivalent saturation current i_{sat} can be derived from (9)-(10)

$$i_{sat} = -(F_{tooth1} + F_{yoke} + F_{tooth2}) \quad (11)$$

Hence, the analytical model based on conformal mapping can be employed to predict the air-gap field in Fig. 5(b). Then, the

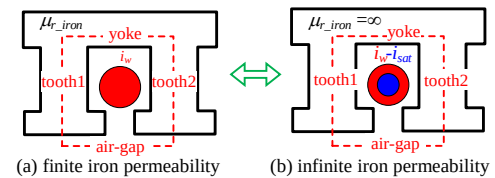


Fig. 5: The Ampere loops to transform the original model with finite iron permeability to the equivalent model with infinite iron permeability.

matrix form of the equivalent saturation current $\mathbf{I}_{\text{sat}}$ can be expressed as

$$\mathbf{I}_{\text{sat}} = \mathbf{A}_{\text{iron}} \mathbf{H}_{\text{r}} \mathbf{L}_{\text{r}} \quad (12)$$

where $\mathbf{H}_{\text{r}}$ and $\mathbf{L}_{\text{r}}$ are the matrix of magnetic field strength and length for the stator and rotor iron. $\mathbf{A}_{\text{iron}}$ is the incidence matrix extracted from the magnetic circuit in Fig. 6.

Then, the simple magnetic circuit is used in the iSAM to obtain the magnetic potential drop in the stator and rotor, as shown in Fig. 6. The flux sources φ_r flowing into the stator or the rotor are calculated from the linear analytical solution based on (6)-(8).

$$\varphi_r = r_i l_{ef} \sum_k [(A_k(i_k, r_{\varphi 2}, \theta_{\varphi 2}) - A_k(i_k, r_{\varphi 1}, \theta_{\varphi 1})] \quad (13)$$

where r_i represents either the radius of the stator bore or the rotor outer surface. l_{ef} is the axial effective length of the SPM motor. The positions $(r_{\varphi 2}, \theta_{\varphi 2})$ and $(r_{\varphi 1}, \theta_{\varphi 1})$ in the Ψ plane of Fig. 4 are conformally mapped to the upper and lower bounds of the integral surface for the flux φ_r in the S plane. Then, the matrix form of φ_r can be derived from (13)

$$\Phi_{\text{r}} = \mathbf{F}_{\text{k}} \mathbf{I}_{\text{k}} = \mathbf{F}_{\text{PM}} \mathbf{I}_{\text{PM}} + \mathbf{F}_{\text{w}} \mathbf{I}_{\text{w}} + \mathbf{F}_{\text{sat}} \mathbf{I}_{\text{sat}} \quad (14)$$

where $\mathbf{F}_{\text{k}}$ represents the relationship between the flux distribution and equivalent current in the PM motor and it is independent of the equivalent current matrix $\mathbf{I}_{\text{k}}$. Φ_{r} is the matrix of $\varphi_{st1}, \varphi_{st2}, \dots, \varphi_{rt1}, \varphi_{rt2}, \dots$. According to Fig. 6, φ_{sy1} is the magnetic flux in the stator yoke, $\varphi_{st1}, \varphi_{st2}$ are the magnetic flux in the stator teeth. Since the slot flux leakage and tooth tip saturation are neglected in the simplified magnetic circuit model, $\varphi_{st1}, \varphi_{st2}$ are equal to the air-gap flux flowing into the stator tooth. $\varphi_{ry1}, \varphi_{ry2}$ are the magnetic flux in the rotor yoke, while $\varphi_{rt1}, \varphi_{rt2}$ represent the air-gap flux flowing into the rotor iron. The reluctance of each segment in the magnetic circuit of Fig. 6 is calculated using the corresponding geometry and iron BH curves. The air-gap flux including $\varphi_{st1}, \varphi_{st2}, \varphi_{rt1}, \varphi_{rt2}$ is derived from the analytical air-gap field solution in Section III(A).

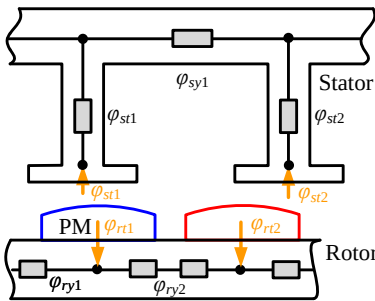


Fig. 6: Simplified iron magnetic circuit of the SPM motor in the iSAM framework.

To accelerate the calculation process, the simplified BH curves are proposed in Fig. 7, which consists of two (Simplified line 1 and 2(2-1, 2-2)) or three linear lines (Simplified line 1, 2-1, and 3). The segmentation of iron BH curve with only two simplified lines helps to obtain the saturation current of iron $\mathbf{I}_{\text{sat}}$ without iterative calculation [25], while maintaining acceptable calculation accuracy. The calculation process is also

shown in the blue block of Fig. 8. Meanwhile, it is possible to employ the third line for severely saturated PM motors, depending on the iron saturation level. When the average flux density in the iron yoke or tooth exceeds B_{imax0} (defined in Fig. 7), it is feasible to employ three linear segments for the iron BH curves. The calculation process for the SPM motor at heavy load can be found in the green block of Fig. 8.

When neglecting iron saturation, the iron flux density can be directly obtained using (14). Then, the simplified line representing the iron BH curve can be selected according to the value of iron flux density neglecting saturation. The matrix of the magnetic field strength $\mathbf{H}_{\text{r}}$ and the iron flux density $\mathbf{B}_{\text{r}}$ can be expressed as

$$\mathbf{H}_{\text{r}} = \mathbf{K}_{\text{sat}} \mathbf{B}_{\text{r}} + \mathbf{H}_0 \quad (15)$$

where $\mathbf{K}_{\text{sat}}$ and $\mathbf{H}_0$ are extracted from Fig. 7. Then, considering $\Phi_{\text{r}} = \mathbf{B}_{\text{r}} \mathbf{S}_{\text{r}}$, the final solution of $\mathbf{I}_{\text{sat}}$ can be obtained from (12), (14)-(15)

$$\mathbf{I}_{\text{sat}} = (\mathbf{E}_0 - \mathbf{F}_{\text{sat}})^{-1} \mathbf{A}_{\text{iron}} \mathbf{L}_{\text{r}} [\mathbf{H}_0 + \mathbf{K}_{\text{sat}} \mathbf{S}_{\text{r}}^{-1} (\mathbf{F}_{\text{PM}} \mathbf{I}_{\text{PM}} + \mathbf{F}_{\text{w}} \mathbf{I}_{\text{w}})] \quad (16)$$

where $\mathbf{S}_{\text{r}}$ is the matrix of the cross-sectional area for the stator and rotor iron. $\mathbf{E}_0$ is the identity matrix.

Then, according to (6)-(8) and (16), the back-EMF can be predicted according to the winding layout and the electromagnetic torque is calculated using Maxwell stress tensor [24].

$$E_{ph} = \frac{d\Psi_{ph}}{dt} = N_c l_{ef} \sum_w \left[\frac{A_{z2}(t)}{dt} - \frac{A_{z1}(t)}{dt} \right] \quad (17)$$

$$T_e = \frac{l_{ef} r^2}{\mu_0} \int_0^{2\pi} B_r B_{\theta} d\theta \quad (18)$$

where N_c is the turns of coil. $A_{z2}(t)$ and $A_{z1}(t)$ are the vector potential in the slot center produced by the equivalent current.

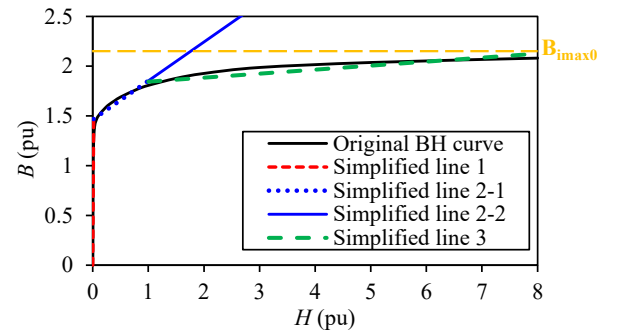


Fig. 7: The simplified BH curve in the iSAM.

C. Solving Process

The solving process of the iSAM is given in Fig. 8. The database in the third step of Fig. 8 can be used to predict the performance of any SPM motors regardless of the geometry variations. The key to predicting the magnetic field in the SPM motor is to transform the eccentric magnetic pole into the equivalent magnet current. Then, the saturation current of iron can be obtained from the combination of the analytical model

and the magnetic circuit. The number of equivalent currents on the magnet surface significantly affects the computational time in the iSAM. Usually, one equivalent magnet current dot can represent the magnetization with a length of 0.5 mm - 1.0 mm while keeping good calculation accuracy and speed. The number of the equivalent magnet current will be discussed in the case study section.

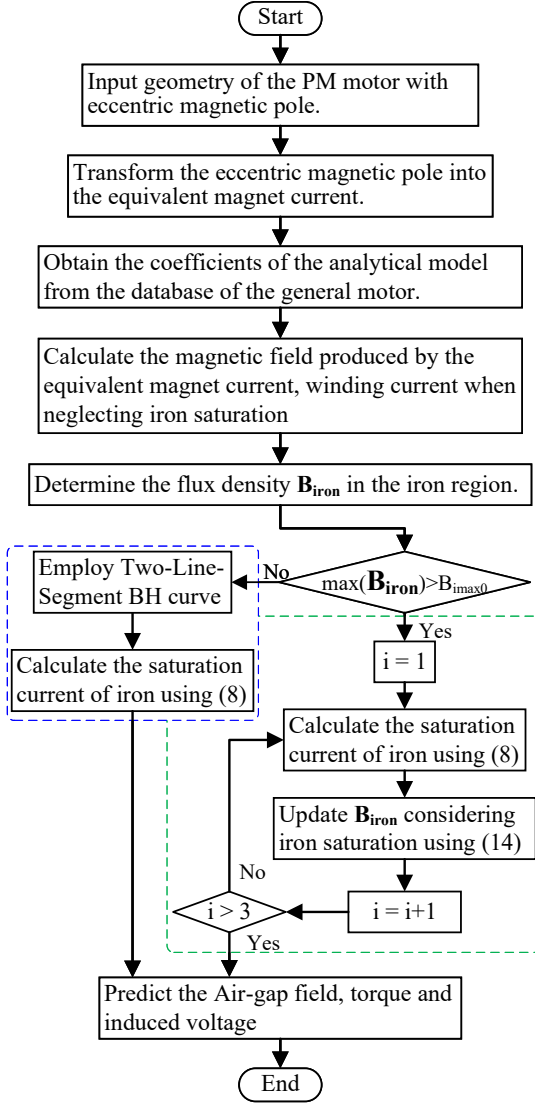


Fig. 8: The solving process of the iSAM.

IV. CASE STUDY

The PM motors with eccentric magnetic poles are developed to validate the proposed method. Both radial and parallel magnetization for PMs are used in the FEM analysis. Fig. 9 shows the flux density distribution of the SPM motor with radially magnetized PMs. As for the geometry of the motor with parallelly magnetized PMs, the full specifications of the motor are given in Table II. Therefore, the detail of the motor cannot be presented, but Fig. 9 can be used as a good representative. Moreover, Fig. 10 shows the test bench of the servo motor with parallelly magnetized PMs. All experiments were conducted at room temperature ($22^\circ\text{C} \pm 1^\circ\text{C}$). The load torque was controlled by a closed-loop dynamometer,

maintaining a fluctuation range below 1%, as measured by the MAGTROL TMB 311/411 torque sensor. The motor voltage was measured using a Tektronix THDP0200 high-voltage differential probe. Each test run lasted less than 30 s, ensuring that the temperature rise of the entire motor was negligible and did not affect the residual flux density (B_r) of the magnets. These procedures ensure that all experimental data is accurate, repeatable, and free from temperature-induced deviations.

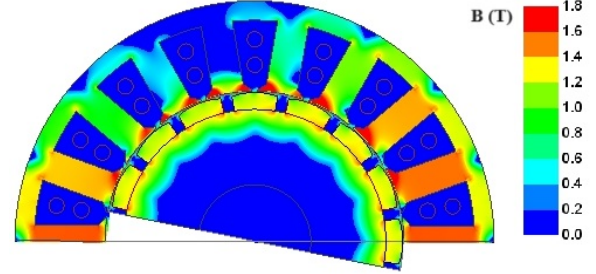


Fig. 9: The flux density distribution of the motor with radially magnetized PMs.

TABLE II: Prototype Specifications

Parameter	Value	Parameter	Value
Rated torque	37 Nm	Rated speed	1200 rpm
Torque constant	2.19 Nm/A	Voltage constant	229.1 V/krpm
Phase resistance	0.114 Ω	Phase inductance	1.21 mH

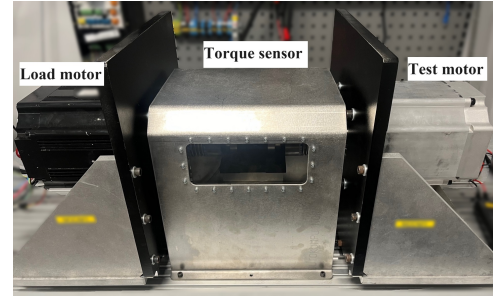


Fig. 10: The test bench of the motor with parallelly magnetized PMs.

A. Calculation accuracy

The radial and tangential components of the on-load air-gap flux density for both motors are predicted using both FEM and iSAM, as shown in Figs. 11-14. They show high accuracy of the iSAM. Only small errors are observed at the peak of the waveform. Then, the open-circuit back-EMF can be obtained for both motors, as shown in Figs. 15-16. iSAM exhibits great accuracy in predicting EMF waveform for both motors. The error of predicted back-EMF amplitude using iSAM is smaller than 3% for the prototype with parallelly magnetized PMs.

Then, the torque is also predicted using iSAM at different input currents, and the analytical calculations are compared with FEM results. The on-load torque waveforms using iSAM and FEM are shown in Figs. 17-20. The average errors of the on-load torque are smaller than 4.3% for both SPM motors. As the experimental flux density on the magnet surface is larger

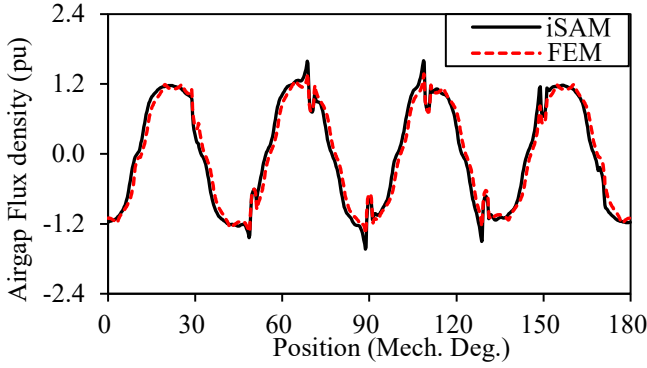


Fig. 11: The radial air-gap flux density of the motor with radially magnetized PMs.

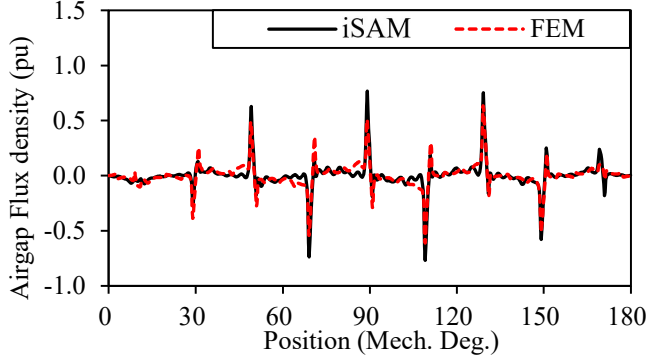


Fig. 12: The tangential air-gap flux density of the motor with radially magnetized PMs.

than the values in the datasheet, the measured torque value is slightly higher than the FEM calculation, making the iSAM torque prediction more accurate than FEM.

To further verify the stability of iSAM under different geometric parameters, heavy-load simulations were conducted for a feasible range of slot opening and air-gap. The variations of average torque with respect to slot-opening width and air-gap length are shown in Figs. 21-22. The average errors of iSAM calculations are 2.7% and 3.6% for radially and parallelly magnetized SPM motors, respectively, demonstrating that the proposed method maintains high accuracy across different geometric conditions. It is noted that when the air-gap and slot-opening become excessively small and the tooth tip saturation cannot be neglected, the maximum errors can reach 9.0% and

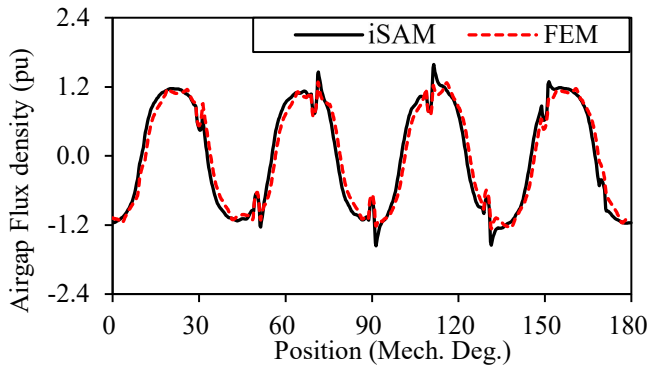


Fig. 13: The radial air-gap flux density of the motor with parallelly magnetized PMs.

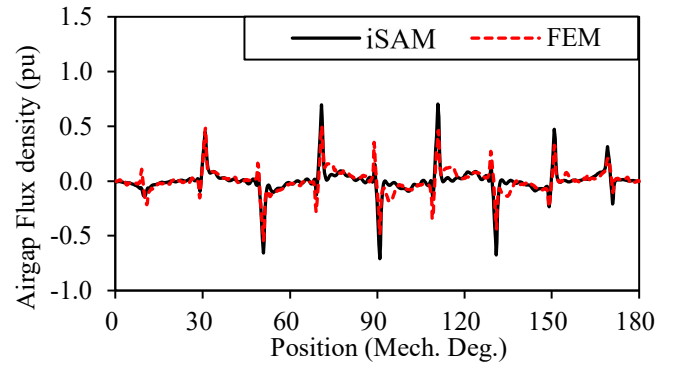


Fig. 14: The tangential air-gap flux density of the motor with parallelly magnetized PMs.

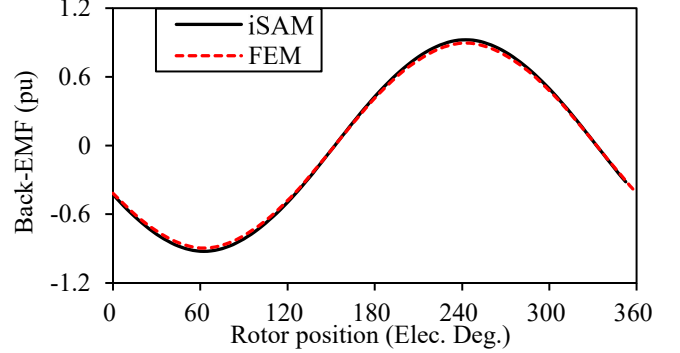


Fig. 15: The open-circuit back-EMF waveform of the motor with radially magnetized PMs.

6.5% in Figs. 21-22, respectively.

B. Calculation time

Tables III-IV give the basic parameter settings using FEM and iSAM. The numbers of nodes and elements in FEM are much larger than those in iSAM. Table V shows the computation time of FEM and iSAM for predicting the motor performance. The open-circuit performance predictions of the SPM motors require over five times the computation time than other cases, as the pre-calculation time is added in the first row of the iSAM predictions. The pre-calculation will get the necessary coefficient in the linear analytical equation for solving the consequent iSAM, which is the most time-consuming process. As for FEM, a similar calculation resource

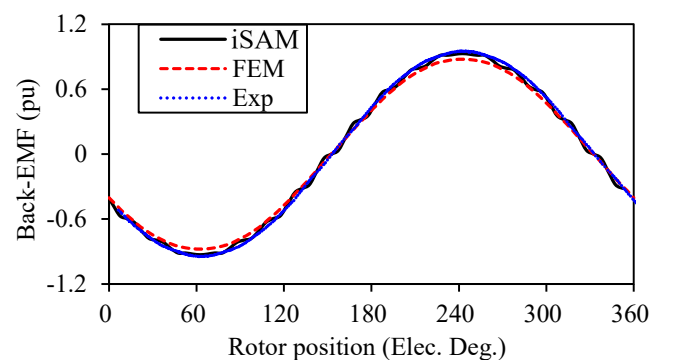


Fig. 16: The open-circuit back-EMF waveform of the motor with parallelly magnetized PMs.

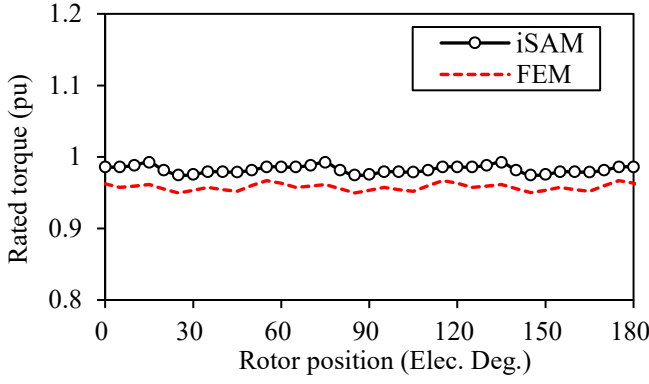


Fig. 17: The torque waveform of the motor with radially magnetized PMs.

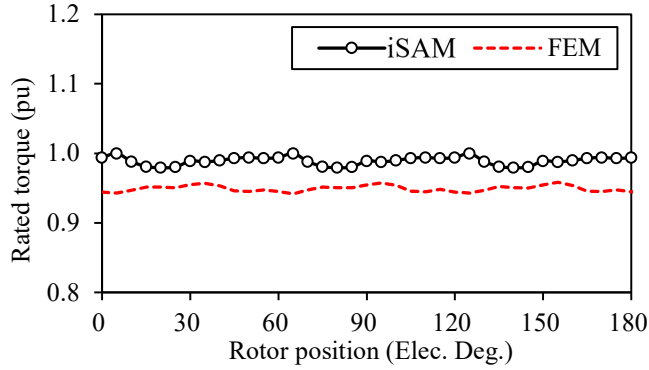


Fig. 18: The torque waveform of the motor with parallelly magnetized PMs.

is consumed at different load conditions, as shown in Table V. It is concluded that the iSAM demonstrates a great advantage when predicting the performance of the SPM motors under different work conditions. Even when the geometry of the motor changed, the iSAM offered approximately a 75% reduction in computational time compared to FEM.

TABLE III: FEM Calculation Parameters for Both motors

Motor type	Mesh Node	Mesh Element	Nonlinear Residual
Radial	89772	43756	0.0001
Parallel	97720	47724	0.0001

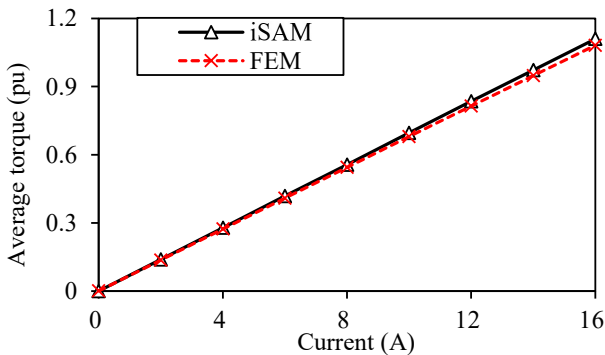


Fig. 19: The average torque and the input current for the lightly loaded motor with radially magnetized PMs.

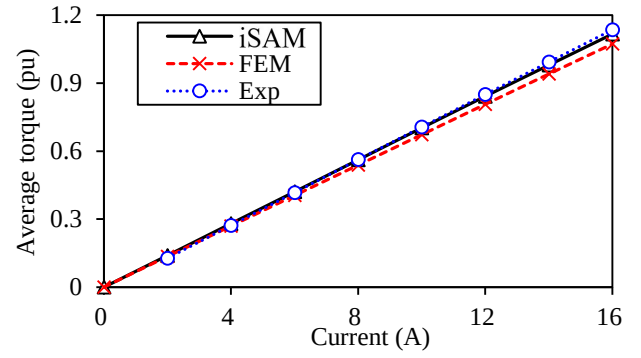


Fig. 20: The average torque and the input current for the lightly loaded motor with parallelly magnetized PMs.

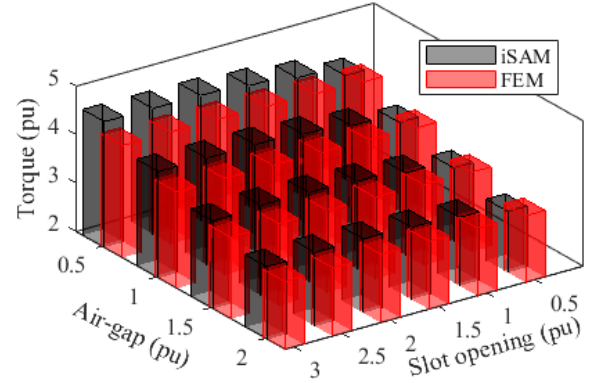


Fig. 21: Variation of average torque with slot-opening width and air-gap length for the heavily loaded motor with radially magnetized PMs.

C. Number of BH curve segmentation

When the motors are operated at heavy load, the iSAM will underestimate the torque with two linear segments for the iron BH curve, as shown in Figs. 23-24. Both the approximated iron BH curve and the absence of an iterative calculation process for determining the working point of the iron are responsible for the underestimation of torque. Therefore, the iSAM with three linear segments for the BH curve approximation is proposed to keep the average torque error smaller than 5% for both motors under heavy load conditions. In addition, three

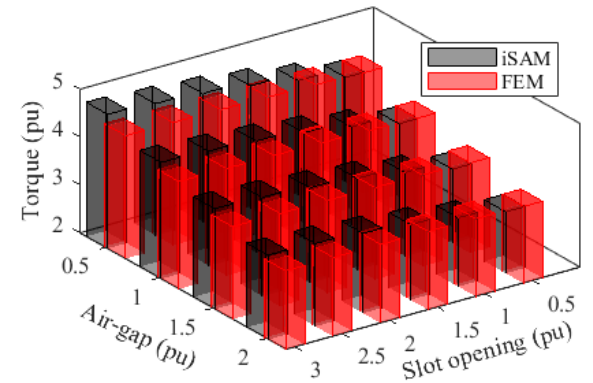


Fig. 22: Variation of average torque with slot-opening width and air-gap length for the heavily loaded motor with parallelly magnetized PMs.

TABLE IV: iSAM Calculation Parameters for Both Motors

Model type	Circuit Node	Circuit Branch	Harmonic order
Radial	34	34	50
Parallel	34	34	50

TABLE V: Computational Time of iSAM and FEM

Current (A)	Radial FEM (s)	Radial iSAM (s)	Parallel FEM (s)	Parallel iSAM (s)
0	501	119	521	141
4	507	21.7	546	20.8
8	515	21.7	552	19.4
12	519	21.7	571	22.1
16	520	19.7	597	24.4

iterations are required to avoid overestimating the equivalent saturation current, as shown in Figs. 23-24. When the input current reaches 30 A, the tooth flux density could reach 1.81 T for the motor with radially magnetized PMs, and 1.79 T for the motor with parallelly magnetized PMs. The iSAM with a two-segment BH curve shows great accuracy for both motors, with errors of 1.9% and 4.9%, respectively. However, the torque errors could increase to 8.7% and 12.1% at 50 A when the tooth flux density reaches 2.02 T and 1.97 T for the motors with radially magnetized PMs and parallelly magnetized PMs, respectively. For such heavy load working conditions, the iSAM with a three-segment BH curve keeps the torque error below 3.6% and 2.6%, respectively, as shown in Figs. 23-24. Nevertheless, the process of approximating iron BH curves and few iterations of determining the equivalent saturation current involve a trade-off between computational efficiency and accuracy when estimating the motor performance.

Subdomain models are effective for preliminary SPM designs, but they typically assume linear iron properties [12]–[16]. Their accuracy is comparable to that of linear FEM solutions, where the iron permeability is fixed at 10^6 . Therefore, linear FEM serves as a fair benchmark representing such analytical models, while also avoiding infeasible parameter dependencies (e.g., the number of harmonics in Fourier expansion) that arise in subdomain formulations. Figs. 23-24 further illustrate that linear FEM, which represents these analytical methods, exhibits significant torque prediction errors at heavy load condition, due to neglecting severe iron saturation.

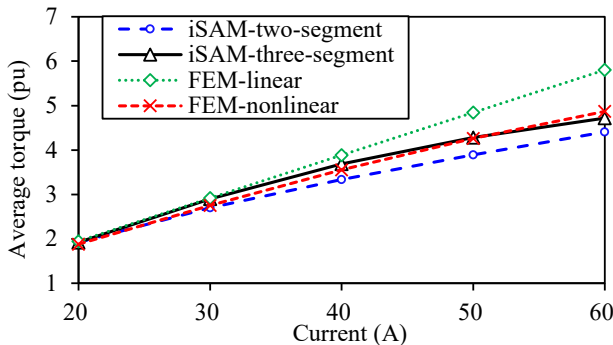


Fig. 23: The average torque and the input current for the heavily loaded motor with radially magnetized PMs.

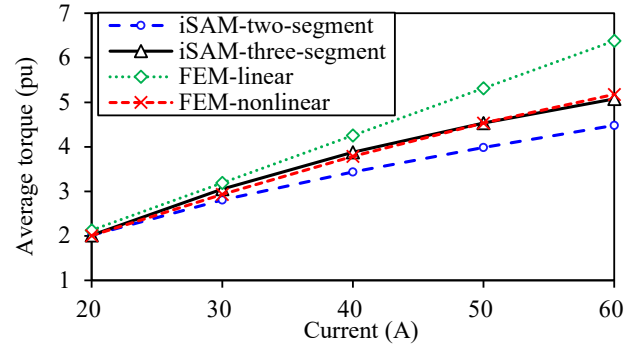


Fig. 24: The average torque and the input current for the heavily loaded motor with parallelly magnetized PMs.

D. PM length per current

The PM length per current significantly influences both the computational accuracy and speed of the proposed iSAM method. It is calculated using equations (1)-(5), as is defined in Section II. Figs. 25-26 illustrate the trade-off between computational time and torque ripple prediction error. From Figs. 25-26, reducing the PM length per current improves accuracy but increases computation time. Meanwhile, a sensitivity analysis of the flux density harmonic with respect to the PM length per current is given in Fig. 27-28 to show the influence of the segmentation current density on the major harmonics of air-gap field. As the fundamental magnetic field harmonic and main harmonic is close to the FEM predictions, the average torque calculation is accurate regardless of the PM length per current. However, when fewer equivalent magnet currents are used in the calculation, the torque ripple errors increase due to the inaccurate representation of the air-gap field harmonics, e.g., the 19th harmonic content in Fig. 27-28. Based on the validation work and practical experience, we found that selecting a segment length in the range of 0.5 mm to 1.0 mm offers a good balance. This range ensures that the torque ripple error remains within acceptable limits (below 3.0%) while avoiding unnecessary computational burden. In this paper, a value of 0.7 mm is chosen for both motors, which falls within this optimal range and results in accurate torque ripple prediction with minimal computation time. Although the relative errors in Fig. 26 appear large due to the small torque ripple magnitude, the absolute torque ripple errors remain below 3.0%, confirming the effectiveness of the selected discretization.

V. CONCLUSION

This paper proposes the iSAM for predicting the SPM motor with eccentric pole-arc shape and different magnetization directions. It introduces the technique of replacing the magnet with an arbitrary magnet shape by the equivalent magnet current. The analytical solution of the air-gap field can be obtained by conformally mapping the SPM motor to the general motor in the database. The iron saturation can be analytically represented using the saturation current of iron, which is extracted from the simple magnetic circuit. The proposed model can accurately predict motor performance with great calculation speed, especially for the different working conditions. The servo PM prototype is investigated and

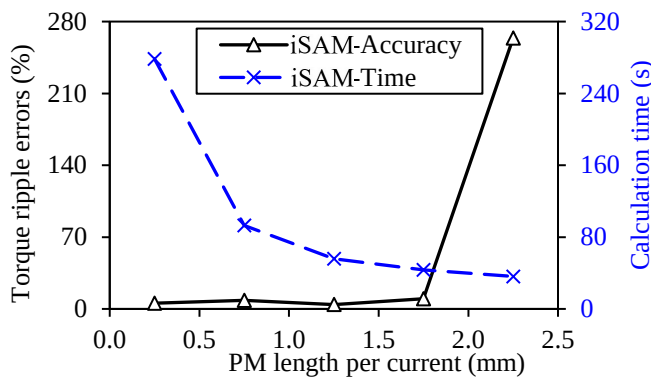


Fig. 25: The variation of the calculation accuracy and speed to PM length per current for the motor with radially magnetized PMs.

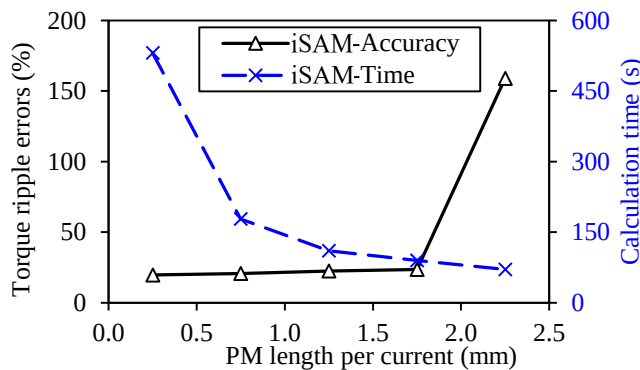


Fig. 26: The variation of the calculation accuracy and speed to PM length per current for the motor with parallelly magnetized PMs.

tested to validate the effectiveness of the iSAM. The high computational efficiency achieved by iSAM is particularly beneficial for iterative optimization workflows such as multi-objective design, torque ripple minimization, and geometry parameter optimization. Since the proposed iSAM does not consider eddy current and end effects, it could lead to low accuracy for high-speed PM motors or axial-flux PM motors. One of our future works will focus on extending the analytical formulation to account for 3D flux leakage using the simple magnetic circuit model and correction coefficients [33], [34].

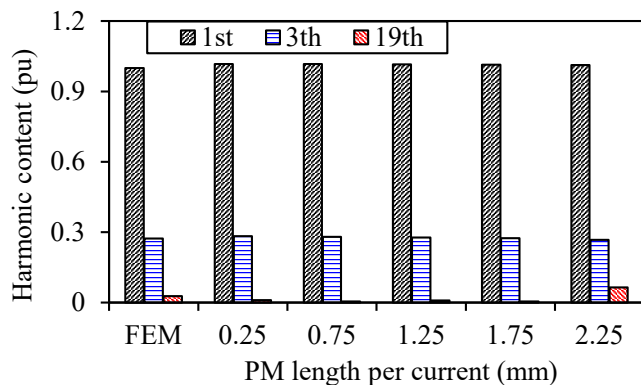


Fig. 27: The air-gap field harmonic content of the motor with radially magnetized PMs at different PM length per current predicted by iSAM.

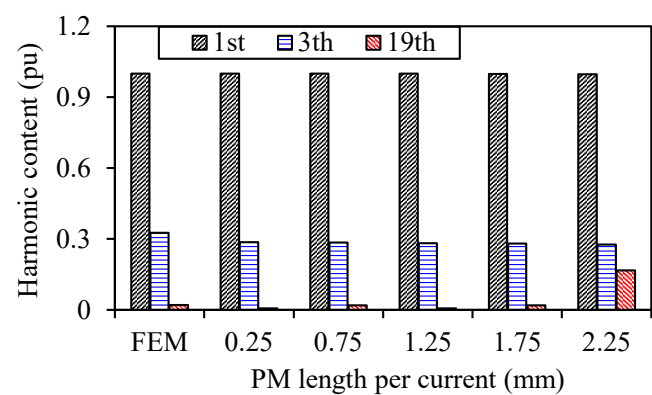


Fig. 28: The air-gap field harmonic content of the motor with parallelly magnetized PMs at different PM length per current predicted by iSAM.

Besides, the influence of thermal behavior is another important aspect to estimate the motor performance. We believe that the proposed iSAM has strong potential to become a more comprehensive model that can handle multi-physical analysis.

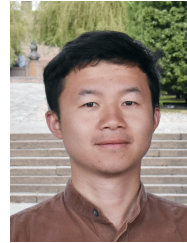
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